

PROJECT DESCRIPTION

– Sriram Chowdary Vundavalli.

TITLE: QUELIFE Animal Shelter Organization.

OBJECTIVE:

QUELIFE is an animal shelter organization that channels all its resources into the selfless undertaking of animal rescue. The entity has centers in seven different states in the United States of America, namely Wisconsin, Illinois, Missouri, Montana, California, Texas, and Nevada. The entity offers a range of essential services intended to provide for animals in need, from giving them shelter to comprehensive care, and facilitating their adoption. They are dedicated to various kinds of domestic animals and see to it that numerous find secure shelter and good care.

A comprehensive database management system (DBMS) needs to be implemented to effectively oversee animal records, adoption processes, staff management, financial sponsorship, and shelter management. The database will provide effective tracking, documentation, and analysis of shelter processes.

Entity sets and attributes:

1. Facility

Facility is a place for housing and caring for animals. The database needs to store information about each facility's location, amenities, and operational details.

- **Facility ID** (Primary Key) – A unique identifier for each facility.
- **Facility Name** – The name of the shelter facility.
- **Location** – The state and city where the facility is located.
- **Number of Play Areas** – Total count of play areas available for animals.
- **Health Care Units** – Number of medical care units available in the facility.
- **Office Space Available** – Indicates whether office space for administrative staff exists (Yes/No).
- **Temperature Control** – Specifies if the facility has heating/air-conditioning (Yes/No).

2. Animal

Animals in the shelter need identification and tracking for medical care, adoption, and shelter management.

- **Animal ID** (Primary Key) – Unique identifier for each animal.
- **Name** – The name of the animal (if applicable).
- **Species** – Type of animal (Dog, Cat, Rabbit, Bird, etc.).
- **Breed** – Specific breed of the animal.

- **Date of Birth** – The estimated or actual birthdate of the animal.
- **Date of Acquisition** – The date when the animal arrived at the shelter.
- **Mode of Acquisition** – How the animal was received (Rescue, Stray, Owner Surrender, etc.).
- **Health Status** – General health condition of the animal (Healthy, Sick, Injured, etc.).
- **Facility ID (FK)** – Foreign Key linking the animal to the facility where it is housed.

3. Adoption

The adoption process records the movement of animals to new homes or people and tracks adoption history.

- **Adoption ID** (Primary Key) – Unique identifier for each adoption.
- **Animal ID (FK)** – Foreign Key linking to the adopted animal.
- **Adopter ID (FK)** – Foreign Key linking to the adopter.
- **Adoption Date** – The date when the adoption was completed.
- **Adoption Fee** – Fee paid by the adopter (if any).
- **Follow-up Status** – Indicates whether a follow-up check was conducted (Pending, Completed, Not Required).

4. Adopter

Individuals or organization who adopt animals must have their contact and eligibility information stored.

- **Adopter ID** (Primary Key) – Unique identifier for each adopter.
- **Name** – The full name of the adopter.
- **Address** – Residential address of the adopter.
- **Phone** – Contact number.
- **Email** – Email address of the adopter.
- **Previous Adoptions** – Number of animals adopted by the adopter in the past.
- **Approval Status** – Whether the adopter is approved for adoption (Pending, Approved, Rejected).

5. Employee

Shelter employees handle animal care, adoptions, and administrative duties.

- **Employee ID** (Primary Key) – Unique identifier for each employee.
- **Name** – Full name of the employee.
- **Role** – Job title or position (Veterinarian, Caretaker, Manager, etc.).
- **Facility ID (FK)** – Foreign Key linking the employee to their assigned facility.

- **Education** – Academic qualifications.
- **Certification** – List of certifications related to animal care or shelter management.
- **Criminal Record** – Boolean (Yes/No) indicating any prior criminal records.
- **Contact Info** – Employee’s phone number or email.

6. Sponsor

Financial sponsors provide donations and financial support for shelter operations and need.

- **Sponsor ID** (Primary Key) – Unique identifier for each sponsor.
- **Name** – Sponsor’s full name or company name.
- **Organization Name** – If the sponsor is part of a business, their company name.
- **Contact Info** – Phone number and/or email address.
- **Donation Amount** – The total amount donated by the sponsor.
- **Sponsorship Type** – Type of sponsorship (One-time, Recurring).

7. Donation

Each financial contribution to the shelter is recorded and categorized for auditing and to make the process transparent.

- **Donation ID** (Primary Key) – Unique identifier for each donation.
- **Sponsor ID (FK)** – Foreign Key linking the donation to the sponsor.
- **Facility ID (FK)** – Foreign Key linking the donation to the receiving facility.
- **Amount** – The donation amount.
- **Donation Date** – Date when the donation was received.
- **Purpose** – Purpose of the donation (Medical, Food, Renovation, General Fund).

8. Medical Record

A log of medical checkups, treatments, and health history for each animal.

- **Record ID** (Primary Key) – Unique identifier for each medical record.
- **Animal ID (FK)** – Foreign Key linking the record to the specific animal.
- **Checkup Date** – The date of the medical checkup.
- **Diagnosis** – Medical condition or diagnosis.
- **Treatment** – Description of treatment provided.
- **Vet Name** – Name of the veterinarian or medical professional.
- **Next Appointment** – The scheduled date for the next medical checkup.

Detailed Relationship Descriptions

1. **Facility houses Animals (One-to-Many)**
 - Each shelter facility provides housing for multiple animals.
 - However, an animal can only be assigned to one facility at a time.
2. **Animal is Adopted by Adopter (One-to-One)**
 - Once an animal is adopted, it is no longer available for adoption.
 - Each adoption record connects one adopter to one adopted animal.
3. **Employee Works in Facility (Many-to-One)**
 - A facility has multiple employees, including caretakers, vets, and managers.
 - Each employee works at only one facility.
4. **Adopter Completes Adoption (One-to-Many)**
 - An adopter can adopt multiple animals over time.
 - Each adoption record connects one animal to one adopter.
5. **Sponsor Funds Facility (Many-to-Many)**
 - Sponsors can donate to one or more shelter locations.
 - A single facility can receive financial support from multiple sponsors.
6. **Facility Receives Donations (One-to-Many)**
 - A facility can receive multiple donations over time.
 - Each donation is assigned to only one facility.
7. **Animal Has Medical Record (One-to-One)**
 - Each animal has a unique medical history recorded in a **Medical Record**.
 - A medical record is linked to a single animal.
8. **Supplier Provides Goods (Many-to-Many)**
 - A supplier may provide food, medical supplies, or other materials to multiple facilities.
 - A facility can receive goods from multiple suppliers.
9. **Employee Manages Adoption (One-to-Many)**
 - Each adoption process is overseen by a shelter employee (e.g., adoption coordinator or shelter manager).
 - An employee can manage multiple adoptions, but each adoption is handled by only one employee

Conclusion

QUELIFE is designed to efficiently manage shelter operations, including animal records, adoptions, employees, sponsors, donations, medical care, and suppliers. The database incorporates **various relationships** such as one-to-one, one-to-many, and many-to-many to ensure a well-structured data model.

This system will enable QUELIFE to computerize animal care, adoption follow-up, funds management, and facility management. The next activity is the creation of the Entity-Relationship (E-R) Diagram, which will show these relationships graphically prior to its implementation in Oracle Database. This project provides a scalable and practical solution to support the shelter's mission effectively.

2) Installation of Oracle SQL Developer.

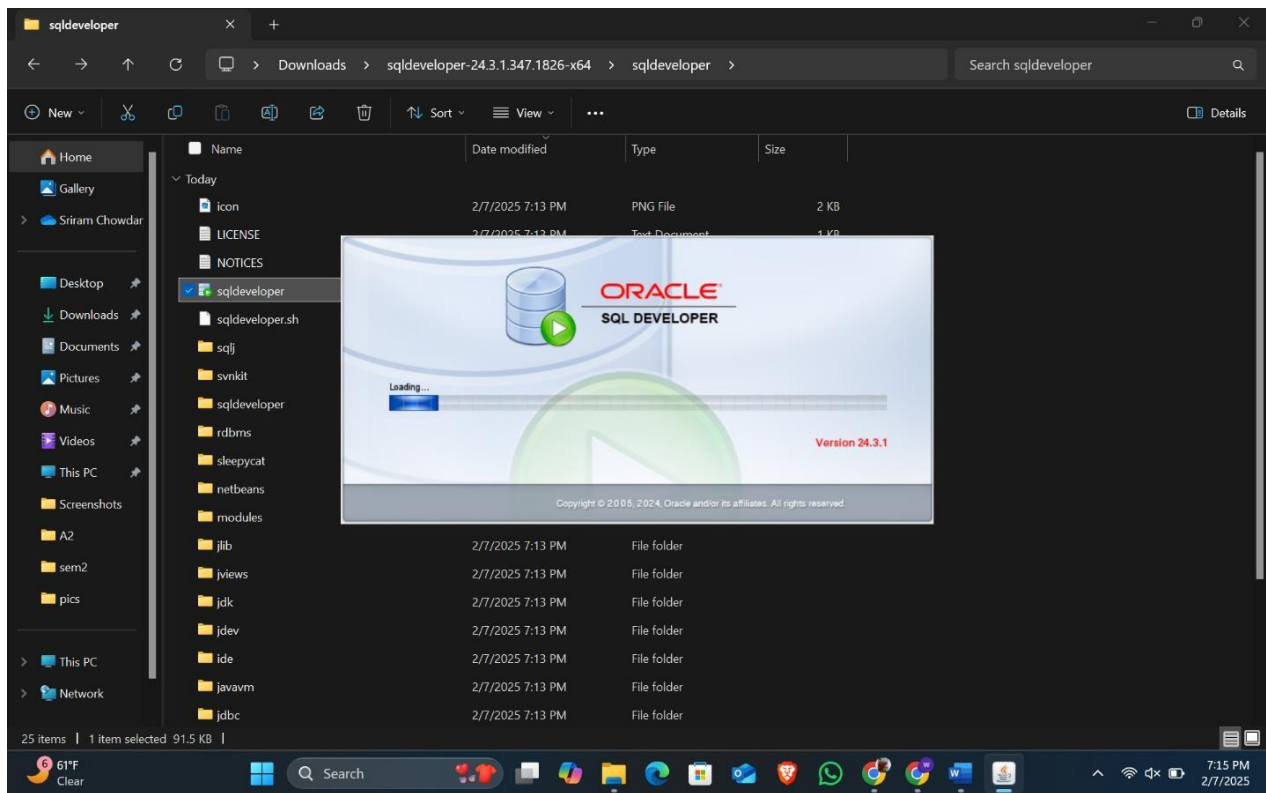
Open chrome and search for sql developer and click on download option.

The screenshot shows a web browser window with the Oracle SQL Developer download page. The browser's address bar shows the URL: `oracle.com/database/sqldeveloper/technologies/download/`. The page features the Oracle logo and navigation links: Products, Industries, Resources, Customers, Partners, Developers, and Company. A search bar and links for 'View Accounts' and 'Contact Sales' are also present.

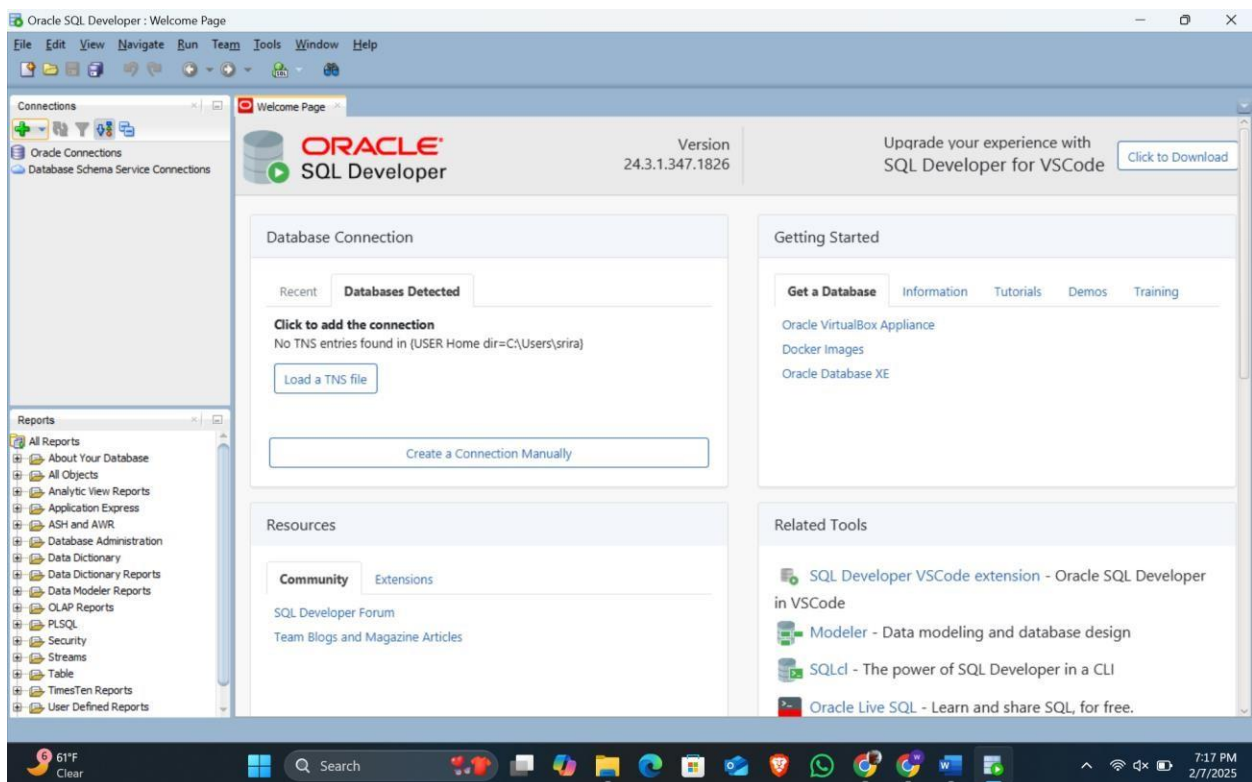
Platform	Download	Notes
SQL Developer Extension for VSCode	Download	- Requires VS Code version 1.82.0 or higher - Available for Win64, OSX (intel/arm), Linux (intel/arm)
Windows 64-bit with JDK 17 included	Download (555 MB)	- MD5: bb9b6abe39f5274b090ce2c7b884f6ff - SHA1: 5295a49db5b39b2640015e535ff1528822a4ba5a Installation Notes
Windows 32-bit/64-bit	Download (557 MB)	- MD5: 7de67da92e557e6cc3f8104c9fad951e - SHA1: 38cab3650bcc67cd8af4f7ca54a6a541 Installation Notes - JDK 17 required

A 'Talk to sales' button is visible in the bottom right corner of the page. The browser's taskbar at the bottom shows the Windows logo, a search bar, and various application icons. The system clock indicates 9:03 PM on 2/7/2025.

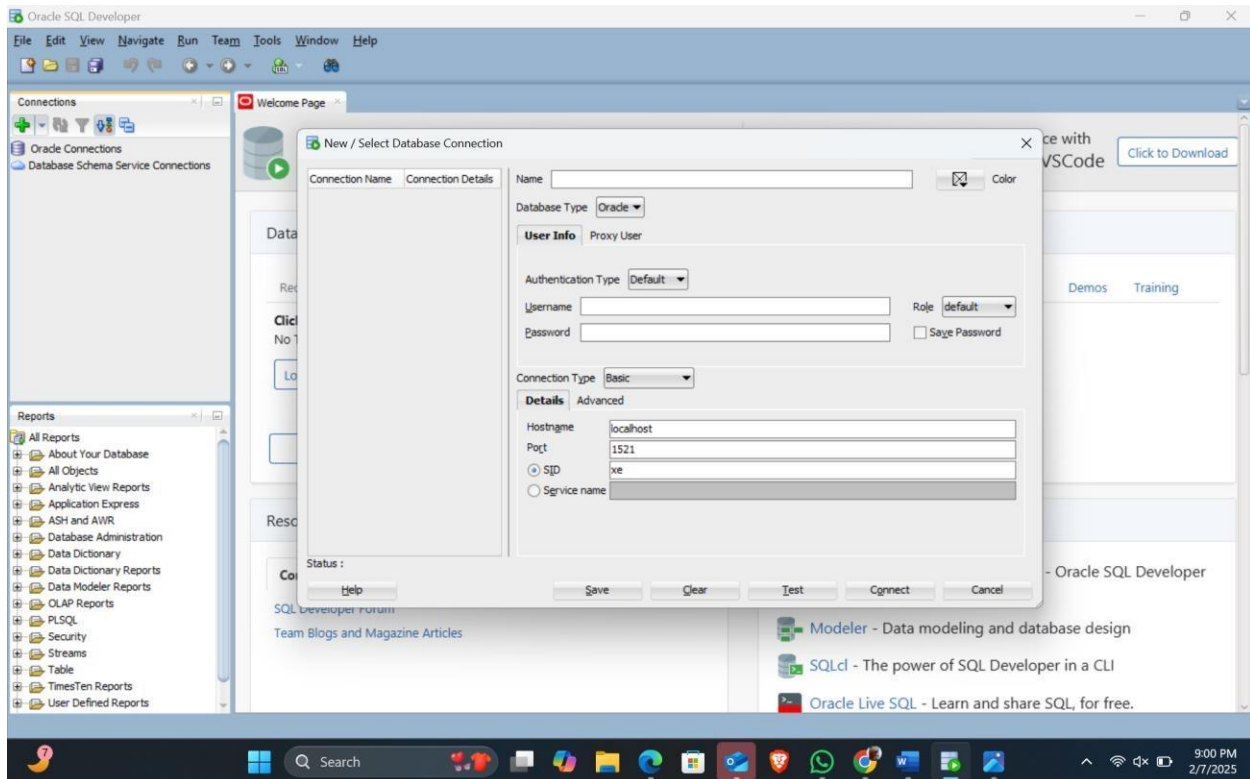
After downloading click on the application and start the installation.



After successful installation we can see the below window



After clicking on the plus button we can set the database connection.



3) Individual Contribution

I have added detailed attributes for the entities **Adoption** and **Adopter** making sure that all the required information like tracking adoptions and adopter details. I have ensured that **Adoption** entity includes attributes such as Adoption ID, Animal ID, Adopter ID, Adoption Date, Adoption Fee and Follow-up status to properly track each and every adoption process. For **Adopter** entity I included attributes Adopter ID, Name, Address, Phone, Email, Previous Adoptions and Approval status to store contact information and the Adopter eligibility information. I have also helped in creating relations like **Adopter Completes Adoption** as a **One-to-Many relationship**, such that one animal is connected to one Adopter and one Adopter to adopt multiple animals. My contribution helps to clearly understand the representation of adoption process in the database system.